

# AKHILESH KUMAR MISHRA

SOFTWARE DEVELOPMENT ENGINEER | MICROSERVICES · DISTRIBUTED SYSTEMS · FULL STACK

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## PROFILE

Software engineer building distributed systems for national e-governance and FinTech platforms — microservice architecture, transaction-critical APIs, event-driven pipelines. Currently 2 million+ users and 500,000+ transactions a day across Java/Spring Boot, Python/FastAPI and Node.js services. Hired and onboarded 4 engineers and moved the team onto Docker-based CI/CD. Most useful when a system is either growing faster than its design or falling over under load.

## CORE TECHNOLOGY

### LANGUAGES

Java, Python, JavaScript (ES6+), TypeScript, SQL, C++, HTML5, CSS3

### BACKEND & ARCHITECTURE

Spring Boot, Django, FastAPI, Node.js, Express.js, React.js, Next.js, microservices, REST APIs, WebSockets

### DATA & MESSAGING

MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, Celery, Hibernate/JPA, indexing & query tuning

### DEVOPS & OBSERVABILITY

Docker, Git, GitHub Actions, CI/CD, Linux, Postman, Jira, Prometheus, Grafana

## EXPERIENCE

12/2023 – PRESENT

New Delhi, India

### Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

- Split monolithic backends into distributed microservices across **Spring Boot**, **FastAPI** and **Node.js**, serving **2 million+ users**.
- Built the transaction-critical endpoints for AEPS, MATM and UPI, sustaining **500,000+ daily transactions**.
- Cut unauthorized access attempts **99.9%** with **JWT**, **RBAC**, device binding and OTP controls.
- Reduced read/write latency **45%** across **MySQL**, **PostgreSQL** and **MongoDB** using sharding, partitioning and query tuning.
- Lifted message throughput **35%** by moving inter-service work onto **Apache Kafka**, **Celery** and **Redis**.
- Stood up **Prometheus** and **Grafana** dashboards and automated container delivery for all **3** payment platforms with **Docker** and **GitHub Actions**.
- Hired and onboarded **4 engineers**, improving team onboarding efficiency **40%**.

06/2023 – 11/2023

Remote, India

### Freelance Software Engineer

Workforce Telemetry & Operations

Real-Time Telemetry & Geofencing Platform

- Engineered a workforce telemetry platform — **React.js** front end, **FastAPI** backend — for **1,000+ field agents**.
- Processed **10,000+ concurrent** GPS pings at **99.9% uptime** through a **FastAPI**, **Celery** and **Redis** ingestion pipeline.
- Built the live map dashboard in **React.js** with **Ant Design**, **30%** quicker to load.

## SELECTED PROJECTS

### Transaction Orchestration & Ledger Platform

Microservice platform orchestrating AEPS, MATM, and UPI settlement with a transactional wallet ledger, retry and reconciliation handling, and Kafka event fan-out.

**STACK:** Spring Boot / FastAPI / Kafka / PostgreSQL / Redis

### MERN FinTech & Commerce Suite

Full-stack applications with JWT-secured authentication, role-based admin dashboards, and versioned REST APIs on React, Express, and MongoDB.

**STACK:** React.js / Node.js / Express.js / MongoDB / JWT / Redux

### Real-Time Telemetry & Geofencing Engine

Workforce geolocation engine ingesting asynchronous GPS streams, rendering live map visualizations, and pushing geofence alerts over WebSockets.

**STACK:** React.js / FastAPI / Celery / Redis / PostgreSQL

### Enterprise Authentication & Authorization Service

Reusable auth service providing registration, password hashing, JWT issuance and rotation, and RBAC enforcement across protected REST routes.

**STACK:** Spring Boot / Spring Security / JWT / MySQL

## EDUCATION

Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 - 2023

## CERTIFICATIONS

Data Structures & Algorithms Certification  
Database Systems & SQL Certification  
Advanced MERN Stack Developer Certification  
IIT Kanpur Cyber Security Certification

## ACHIEVEMENTS

- Graduated with First Class Honors, maintaining top academic rank.
- Earned a 4-Star Gold Badge in SQL on HackerRank.